Difference Equation Problems: MATLAB Exploration

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Set Up Workspace

```
close all
clear

[Song,fs] = audioread('EgmontOverFin.wav');

imp = [1, zeros(1,99)];
lowcos = cos(400*2*pi*(0:199)/fs);
highcos = cos(10000*2*pi*(0:199)/fs);

% Difference equation coefficients System 1
A1 = [1  -2.742912767603523   2.541050804003343  -0.793339740582830];
B1 = [0.014730524709542  -0.012331376801047  -0.012331376801047   0.014730524709542];

% Difference equation coefficients System 2
A2 = [1 -2.124929452958793   1.608710001890780  -0.395359893970549];
B2 = [0.64890416134889  -1.91559551306117   1.91559551306117  -0.64890416134889];
```

Process Song

```
Song1 = filter(B1,A1,Song);
Song2 = filter(B2,A2,Song);

soundsc(Song,fs)  % original song
pause   % hit any key in command window to continue

soundsc(Song1,fs)  % song through system 1
pause   % hit any key in command window to continue

soundsc(Song2,fs)  % song through system 2
```

Roots of the Characteristic Equation

```
'System 1 characteristic equation root magnitudes'
abs(roots(A1))
'System 2 characteristic equation root magnitudes'
abs(roots(A2))
```

```
ans =

System 1 characteristic equation root magnitudes


ans =

   0.951603139881574
   0.951603139881574
   0.876087486413428


ans =

System 2 characteristic equation root magnitudes


ans =

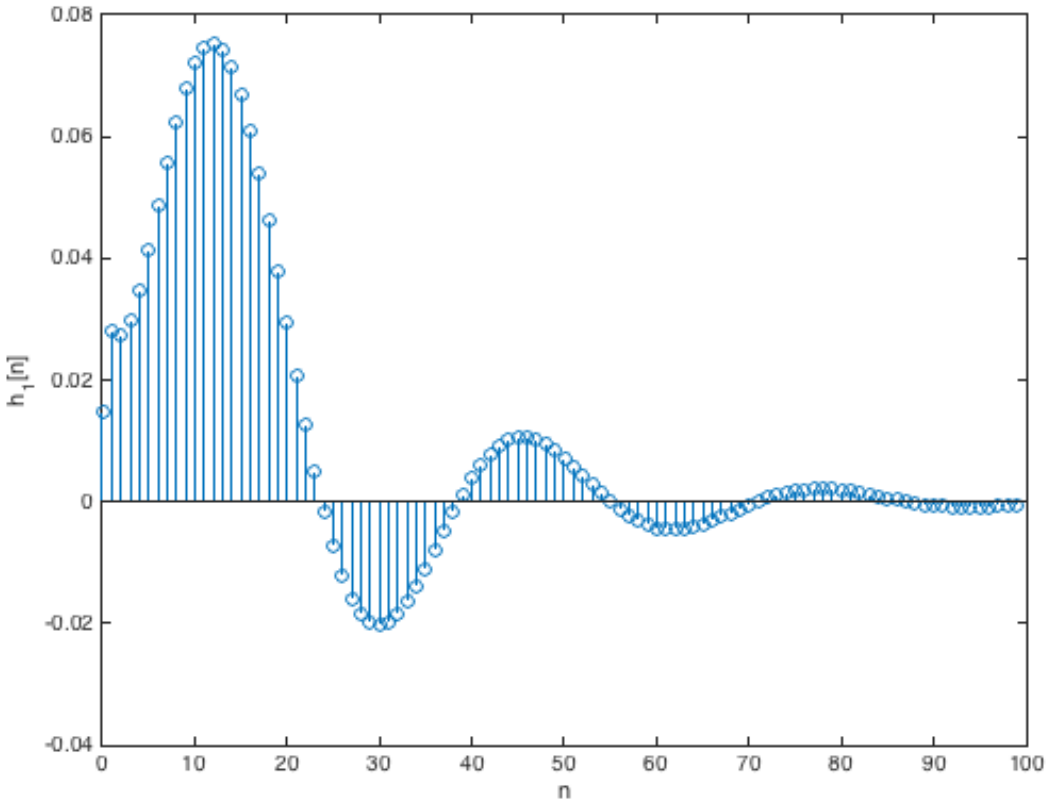
   0.899593511579838
   0.899593511579838
   0.488539836621235
```

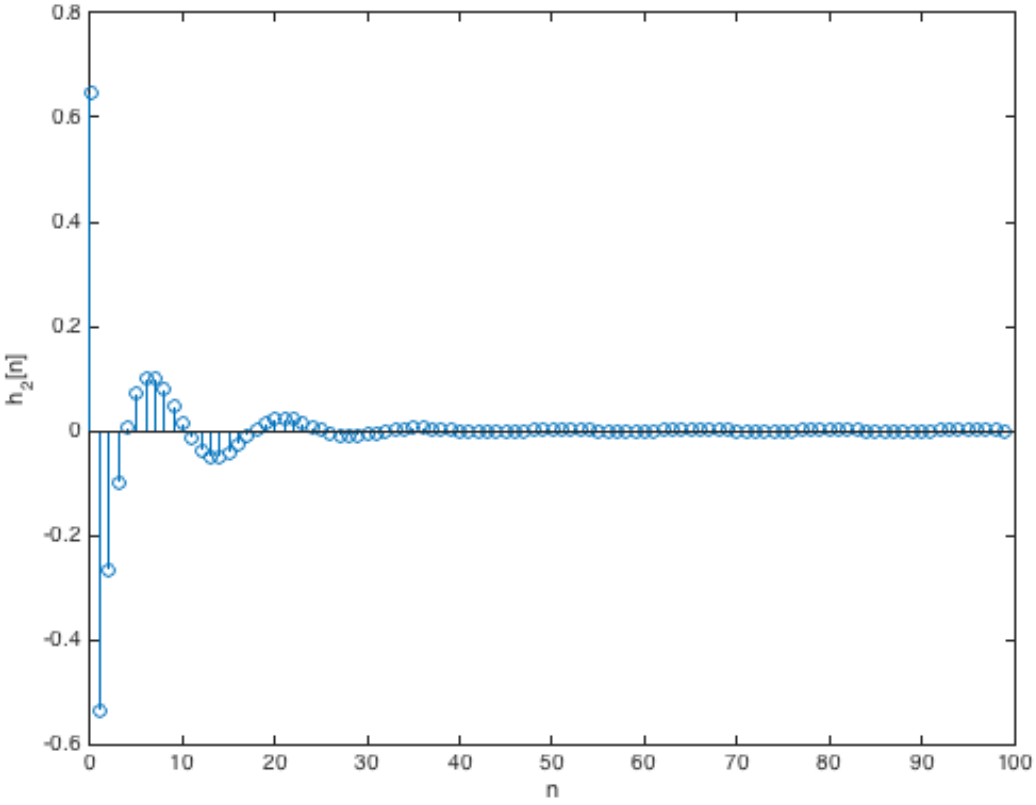
Impulse Responses of the Two Systems

```
h1 = filter(B1,A1,imp);
h2 = filter(B2,A2,imp);

figure
stem(0:99,h1)
xlabel('n')
ylabel('h_1[n]')

figure
stem(0:99,h2)
xlabel('n')
ylabel('h_2[n]')
```





Output for Low and High Frequency Cosine Inputs

```
cl1 = filter(B1,A1,lowcos);
ch1 = filter(B1,A1,highcos);

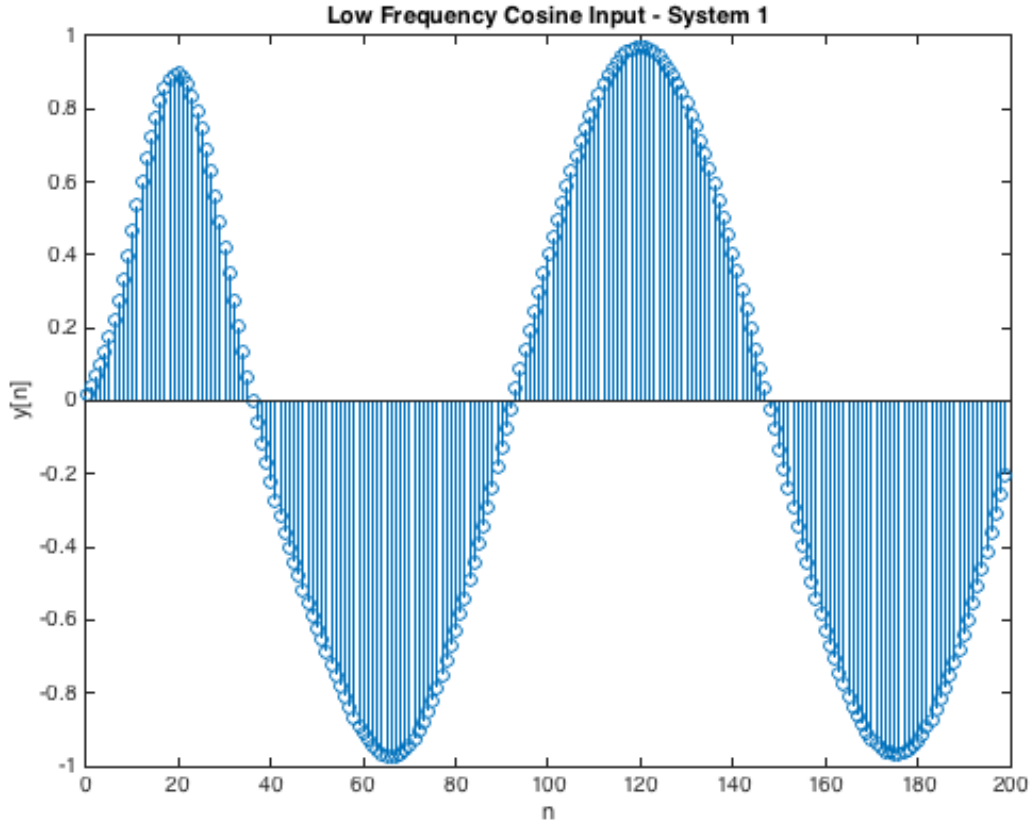
cl2 = filter(B2,A2,lowcos);
ch2 = filter(B2,A2,highcos);

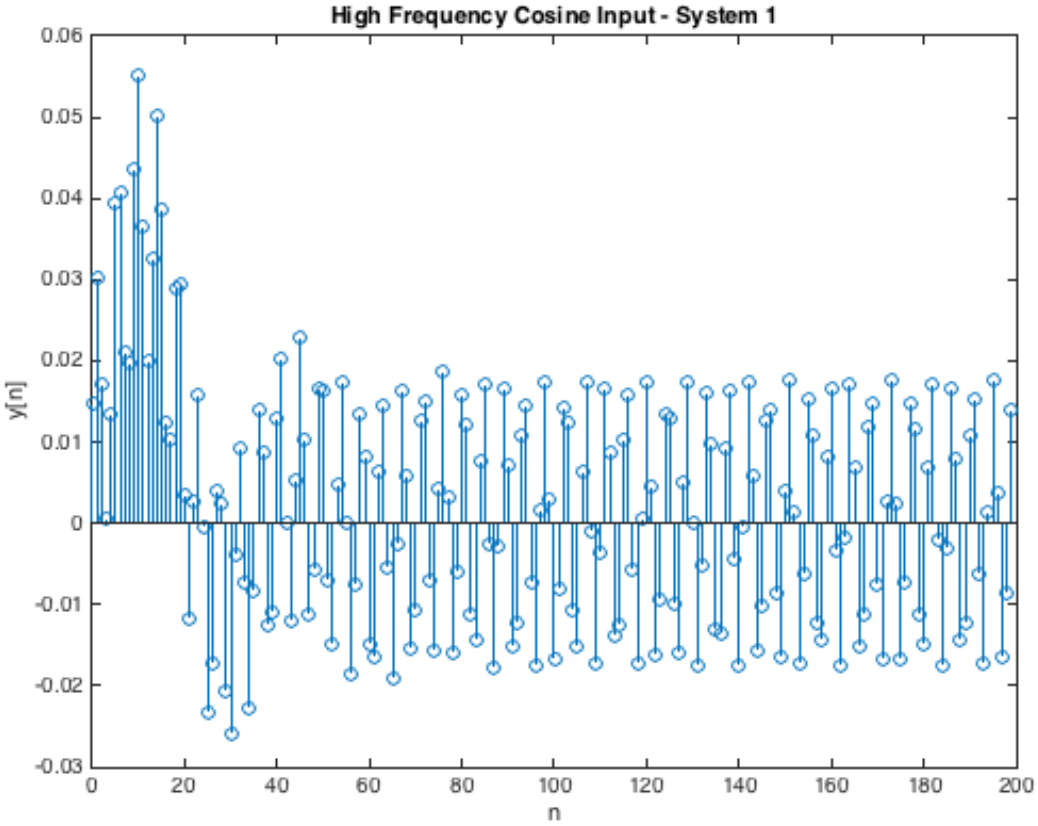
figure
stem(0:199,cl1)
xlabel('n')
ylabel('y[n]')
title('Low Frequency Cosine Input - System 1')

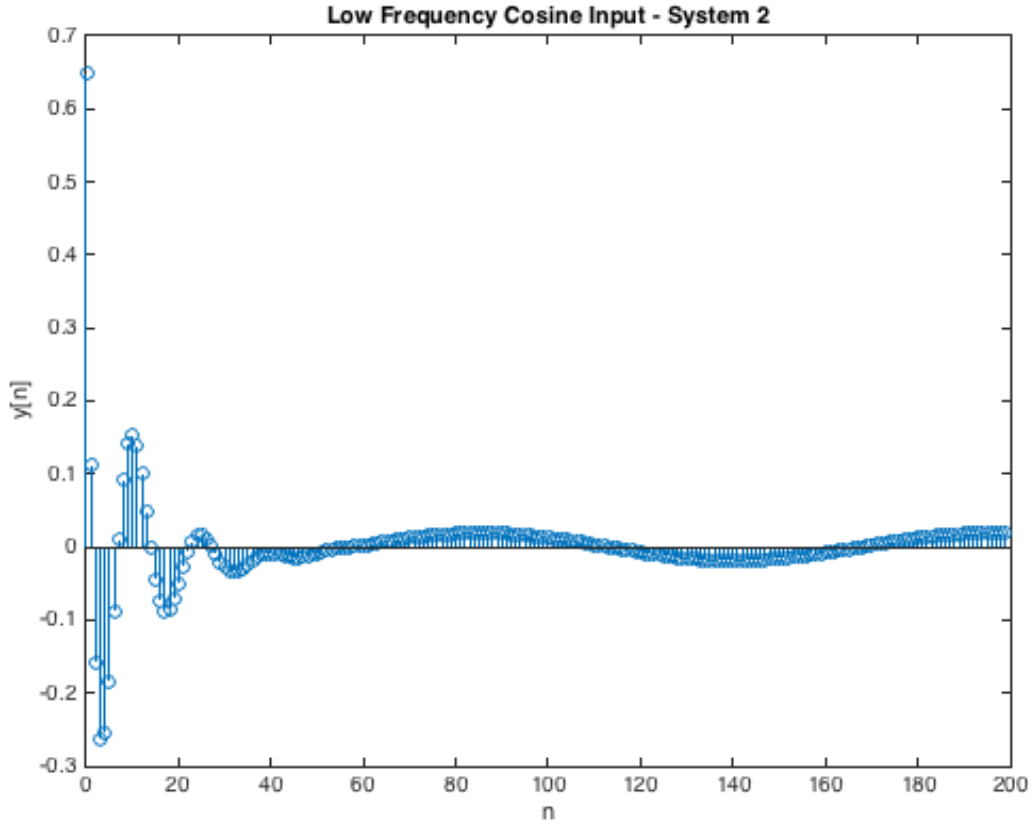
figure
stem(0:199,ch1)
xlabel('n')
ylabel('y[n]')
title('High Frequency Cosine Input - System 1')

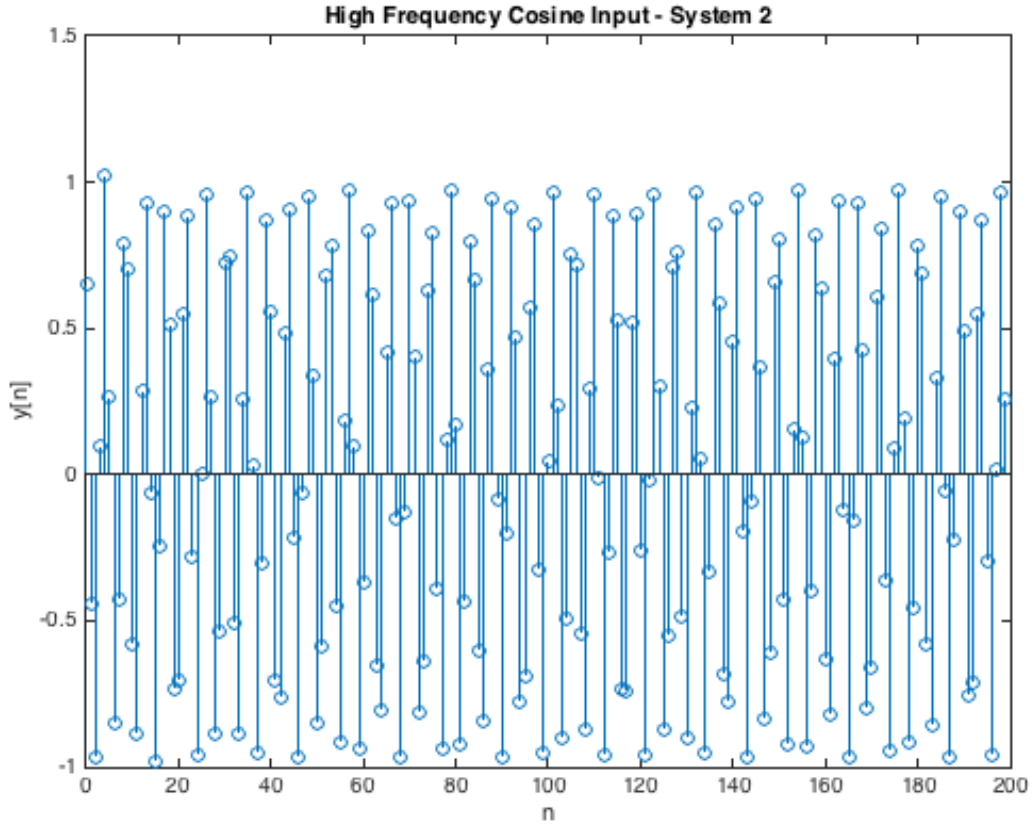
figure
stem(0:199,cl2)
xlabel('n')
ylabel('y[n]')
title('Low Frequency Cosine Input - System 2')

figure
stem(0:199,ch2)
xlabel('n')
ylabel('y[n]')
title('High Frequency Cosine Input - System 2')
```


Low Frequency Cosine Input - System 1
y[n]
n
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
0
20
40
60
80
100
120
140
160
180
200


High Frequency Cosine Input - System 1
y[n]
n
0.06
0.05
0.04
0.03
0.02
0.01
0
-0.01
-0.02
-0.03
0
20
40
60
80
100
120
140
160
180
200


Low Frequency Cosine Input - System 2
y[n]
n
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
-0.1
-0.2
-0.3
0
20
40
60
80
100
120
140
160
180
200


High Frequency Cosine Input - System 2
y[n]
n
1.5
1
0.5
0
-0.5
-1
0
20
40
60
80
100
120
140
160
180
200

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